

Program : Diploma in Communication and Computer Networking	
Course Code : 6327	Course Title: Python Programming
Semester : 6	Credits: 2.5
Course Category: Programming core course	
Periods per week: 4 (L: 0 T: 1 P: 3)	Periods per semester: 45

Course Objectives:

- To write, test, and debug simple Python programs .
- To implement Python programs with conditionals and loops
- Discover the use of supported data structures like lists, dictionaries and tuple in Python

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Python Programming concepts	1008	Introduction to IT Systems lab	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	To write, test, and debug simple Python programs .	12	Applying
CO2	To implement Python programs with conditionals and loops	12	Applying
CO3	Use Python lists, tuples, dictionaries for representing compound data	13	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	3	3	2			
CO2	2	3	3	2			
CO3	2	3	3	2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To write, test, and debug simple Python programs.		
M1.01	Write simple Python program to display message on screen	3	Applying
M1.02	Write simple Python program using operators: a) Arithmetic Operators b) Logical Operators c) Bitwise Operators	4	Applying
M1.03	Fam Write a program to demonstrate basic data types in python	5	Applying
CO2	To implement Python programs with conditionals and loops		
M2.01	Write simple Python program to demonstrate use of conditional statements: a) 'if' statement b) 'if ... else' statement c) Nested 'if' statement	6	Applying
M2.01	Write python program to demonstrate use of loopin statements: a) 'while' loop b) 'for' loop	6	Applying
	Series Test– I	2	
CO3	Use Python lists, tuples, dictionaries for representing compound data		
M3.01	Write python program to perform following operations on Lists: a) Create list	5	Applying

	b) Access list c) Update list (Add item, Remove item) d) Delete list		
M3.02	Write python program to perform following operations on Dictionaries: a) Create Dictionary b) Access Dictionary elements c) Update Dictionary d) Looping through Dictionary	5	Applying
M3.03	Write python program to perform following operations on Tuples: e) Create Tuple f) Access Tuple g) Update Tuplet (Add item, Remove item) h) Delete Tuple	3	Applying
	Series Test– II	2	
	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.

Minimum of 6 experiments (excluding open ended experiments/microprojects) shall be performed.

Compulsory for CIA

The CIA shall be arranged in fourth semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
R1	Python Crash Course: A Hands-On, Project-Based Introduction to Programming by EricMatthes
R2	Python Programming: An Introduction to Computer Science 2nd Edition by John Zelle
R3	Kenneth A Lambert., Fundamentals of Python : First Programs, 2/e, Cengage Publishing, 2016
R4	Harsh Bhasin, Problem Solving and Programming: First Edition, 2022
R5	Zed A Shaw, Learn Python 3 The Hard Way, Addison-Wesley, 2017
R6	Allen B. Downey, Think Python: How to Think Like a Computer Scientist, 2/e, Shroff, 2016
R7	Michael Urban and Joel Murach, Python Programming, Shroff/Murach, 2016

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/python/